

Supporting Information

TAMR-DTI: Target-Aware Modulation and Mamba-Based Representation Refinement for Drug-Target Interaction Prediction

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This file is submitted as Supporting Information and is intended to be read together with the main manuscript.

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1 Data, Preprocessing, and Reproducibility Details

This section preserves and expands the data and implementation information from the previously prepared Supporting Information. It is included so that the new equation guide and standalone artwork do not remove any reproducibility details from the earlier SI version.

1.1 Dataset Sources and Processed Splits

We evaluate four public drug-target interaction benchmarks: BindingDB, BioSNAP, Human, and C. elegans. Each sample contains a drug SMILES string, a protein sequence, and a binary interaction label. The statistics below are computed from the processed training, validation, and test split files used by the experiments. The split files are fixed across repeated runs; the seed is varied for model initialization and training stochasticity.

Table S1: Processed dataset statistics used by the manuscript.

Dataset	Drugs	Proteins	Interactions	Positive	Negative	Positive rate
BindingDB	14,643	2,623	49,199	20,674	28,525	42.02%
BioSNAP	4,505	2,181	27,464	13,830	13,634	50.36%
Human	2,726	2,001	5,997	2,633	3,364	43.91%
C. elegans	1,767	1,876	7,786	3,893	3,893	50.00%

Table S2: Training-split class counts and positive-class weights.

Dataset	Training positives	Training negatives	Positive weight N_-/N_+
BindingDB	14,583	19,856	1.3616
BioSNAP	9,684	9,540	0.9851
Human	1,846	2,351	1.2736
C. elegans	2,727	2,723	0.9985

Data access points are listed in the manuscript Data Availability Statement: BindingDB, BioSNAP, and the Human/C. elegans benchmark files in the TransformerCPI repository.

1.2 Input Processing and Feature-Cache Schema

SMILES strings are encoded by a ChemBERTa hidden-state cache and converted to DGL molecular graphs. Protein sequences are encoded by a ProtBERT hidden-state cache. The data loader validates cache shapes and retains explicit masks for protein residues and invalid conformer slots. The main model uses the vanilla cache variant, `drug3d_features.pt`.

RDKit ETKDGv3 generates up to eight conformers. The extraction routine uses random seed 42, RMSD pruning threshold 0.5, and one embedding thread. UFF optimization is attempted for at most 100 iterations when parameters are available. Molecules with more than 160 heavy atoms are skipped by the 3D extraction guard. Protein sequences are upper-cased after whitespace removal; U, Z, O, and B are replaced with X before token encoding. The model-side protein length is padded or masked to 128. The feature-extraction schema version is 3.

Table S3: Feature-cache tensors and dimensions.

Stream	File or tensor	Shape	Processing and use
Drug 1D	<code>smiles_features.pt</code>	354×128	ChemBERTa token hidden states after adaptive pooling.
Drug graph	DGL graph	Up to 290×75	Canonical atom/bond features; self-loops enabled.
Drug 3D	<code>drug3d_features.pt</code>	$K \times 64 \times 128$	Per-conformer atom features; $K = 8$ in the main model.
Coordinates	3D conformer coordinates	$K \times 64 \times 3$	Coordinates consumed by the EGNN branch.
Conformer meta-data	<code>conf_mask</code> , energy	K , K	Validity mask and UFF energy for target-aware scoring.
Protein	<code>protein_features.pt</code>	128×1024	ProtBERT residue features after pooling; residue mask retained.

1.3 Complete Architecture Configuration

The model uses a 128-dimensional shared latent space. Protein context first conditions target-aware conformer scoring. The resulting drug representation provides a ligand context for FiLM modulation of protein tokens. A bidirectional protein-side Mamba block refines the protein sequence before one-layer BiIntention cross-modal matching.

Table S4: Main TAMR-DTI architecture configuration.

Component	Setting
Drug graph encoder	Input node dimension 75; hidden dimensions [128, 128, 128]; maximum graph length 290.
Target-aware conformer encoder	EGNN dimension 128; $K = 8$; 64 atoms per conformer; protein-conditioned score input; energy projection included.
Drug fusion	Graph and conformer tokens form 354 structural slots; layer-normalized fusion with a learned sigmoid gate initialized at bias -2.0 .
Protein encoder	1024-dimensional ProtBERT features projected to 128; three ProteinACmix layers; kernels 3, 6, 9; eight heads; ligand FiLM scale 0.05.
Protein Mamba	Protein tokens only; bidirectional; $d_{\text{state}} = 16$; $d_{\text{conv}} = 4$; expansion 2; dropout 0.1; Mamba-1 backend; gate bias -2.0 .
BiIntention	One layer; eight heads; embedding dimension 128; bidirectional drug-protein intention updates.
Prediction head	Four-layer MLP; input 256; hidden 256; intermediate output 128; binary logit output.

1.4 Training, Loss, and Evaluation

For completeness, the threshold-dependent metrics used in the manuscript are defined below. The threshold τ is applied to the predicted probability.

$$\text{Accuracy}(\tau) = \frac{\text{TP}(\tau) + \text{TN}(\tau)}{\text{TP}(\tau) + \text{TN}(\tau) + \text{FP}(\tau) + \text{FN}(\tau)}, \quad (\text{S1})$$

$$\text{Precision}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FP}(\tau)}, \quad (\text{S2})$$

$$\text{Recall}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}, \quad (\text{S3})$$

$$\text{F1}(\tau) = \frac{2 \text{Precision}(\tau) \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)}. \quad (\text{S4})$$

Table S5: Training and evaluation settings.

Setting	Value
Optimizer	Adam
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Maximum epochs	100
Batch and accumulation	16 per GPU; gradient accumulation 4 steps
Gradient clipping	Maximum norm 1.0
Scheduler	Plateau; factor 0.5; patience 5; minimum learning rate 1×10^{-6}
Early stopping	Patience 10; minimum improvement 1×10^{-4}
Checkpoint	Highest validation AUROC
Runtime	Python 3.9; PyTorch, DeepSpeed ZeRO-2, DGL/DGLLife, RDKit, Transformers, and mamba-ssm; eight Tesla P100 GPUs.

The validation threshold is searched over 0.01, 0.02, $\dots$, 0.99 and then applied unchanged to the held-out test split. Fixed-threshold results use $\tau = 0.5$.

1.5 Main Run Matrix

All four datasets use the same main architecture and five fixed seeds, 42 through 46. Configuration identifiers correspond to files under `configs/experiments/`.

Table S6: Main repeated-seed configuration index.

Dataset	Seed	Configuration
BindingDB	42	stage2_main_01
BindingDB	43	stage2_main_02
BindingDB	44	stage2_main_03
BindingDB	45	stage2_main_04
BindingDB	46	stage2_main_05
BioSNAP	42	stage2_main_06
BioSNAP	43	stage2_main_07
BioSNAP	44	stage2_main_08
BioSNAP	45	stage2_main_09
BioSNAP	46	stage2_main_10
Human	42	stage2_main_11
Human	43	stage2_main_12
Human	44	stage2_main_13
Human	45	stage2_main_14
Human	46	stage2_main_15
C. elegans	42	stage2_main_16
C. elegans	43	stage2_main_17
C. elegans	44	stage2_main_18
C. elegans	45	stage2_main_19
C. elegans	46	stage2_main_20

1.6 Expanded Performance and Diagnostic Results

The standalone tables in Section 3 reproduce the main AUROC/AUPRC comparison, the repeated-run summary, the BioSNAP module ablation, and the conformer-count diagnostic. All diagnostics use BioSNAP, seed 42, the same fixed split, validation-AUROC checkpoint selection, and fixed-threshold evaluation. They are single-seed mechanism diagnostics rather than repeated-run significance tests.

1.7 Conformer-Weight and Mamba Interpretability

The analysis uses all 5,493 BioSNAP test pairs from the seed-42 target-aware model. For each sample, conformer weights are sorted in descending order, and prediction groups use validation threshold 0.62. Normalized conformer entropy is calculated as

$$\bar{H}_i = -\frac{1}{\log K_i} \sum_{k=1}^{K_i} w_{ik} \log(w_{ik} + \epsilon), \quad (\text{S5})$$

where K_i is the number of valid conformers for sample i and ϵ is a numerical-stability constant.

Table S7: Population-level interpretability statistics.

Group	n	Top-1 weight	Entropy	Mamba refine mean
All test pairs	5,493	0.5975	0.4799	0.094526
True positive	2,175	0.5910 ± 0.2341	0.4876 ± 0.2545	0.093963
True negative	2,328	0.5974 ± 0.2311	0.4808 ± 0.2517	0.095077
False positive	417	0.5857 ± 0.2287	0.4834 ± 0.2468	0.093675
False negative	573	0.6309 ± 0.2401	0.4446 ± 0.2651	0.095047

Table S8: Median sorted conformer weights by prediction group.

Group	Rank 1	Rank 2	Rank 3	Rank 4
True positive	0.5320	0.2479	0.0946	0.0121
True negative	0.5288	0.2457	0.0964	0.0137
False positive	0.5192	0.2579	0.1036	0.0132
False negative	0.5804	0.2317	0.0756	0.0053

Table S9: Auxiliary modulation statistics from the interpretability analysis.

Quantity	Mean
Mamba gate	0.132949
Conformer top-1 margin	0.365227
Protein FiLM layer 1 strength	0.020351
Protein FiLM layer 2 strength	0.013285
Protein FiLM layer 3 strength	0.012323

The sample-level analysis file is `outputs/interpretability/biosnap_seed42/biosnap_seed42_interpretability_samples.csv`. It records prediction groups, threshold, conformer entropy, top-1 weight, conformer margin, Mamba summaries, eight conformer weights, and layer-wise FiLM statistics.

1.8 Reproducibility and Availability

The implementation, preprocessing code, and experiment configuration files are available at <https://github.com/leleleocc/TAMR-DTI>. Model code is under `src/`, feature extraction is under `scripts/pre_extract.py`, and the run configurations listed in Table S6 are under

`configs/experiments/`. To reproduce a run, obtain the public benchmark files, construct the fixed split directory with `train.csv`, `val.csv`, and `test.csv`, build the ChemBERTa, ProtBERT, graph, and 3D caches with the settings above, and launch the corresponding configuration. Select the checkpoint by validation AUROC and evaluate the held-out test set under the metric conventions in this SI and the manuscript.

2 Equation and Notation Guide

This section provides an equation-by-equation reading guide for Section 3 (Methodology) of the main manuscript. The equation identifiers in the first column are the labels used in the manuscript source. The table is deliberately redundant with the prose description so that every tensor, mask, gate, and aggregation operation can be checked without searching through the main text.

2.1 Global conventions and dimensions

Bold upper-case symbols denote matrices or token sequences, bold lower-case symbols denote vectors, and plain lower-case symbols denote scalars. A batch dimension is omitted unless it is needed for interpretation. The principal dimensions of the reported configuration are listed below.

Table S10: Global dimensions and fixed configuration used by TAMR-DTI.

Symbol or setting	Value	Interpretation
d	128	Shared hidden dimension for drug and protein tokens.
N_g	290	Padded graph-token count.
N_c	64	Resampled atom-token count per conformer.
K	8	Number of conformer slots in the main model.
L_d	354	Drug-token length, $N_g + N_c$.
L_p	Variable	Protein-token length after padding and masking.
d_c	Backbone-dependent	ChemBERTa hidden size before pooling.
H	8	Number of heads in the BiIntention module.
ProteinACmix layers	3	Ligand-conditioned protein encoder depth.
BiIntention layers	1	Pairwise interaction depth in the main model.

The mask $m_j \in \{0, 1\}$ indicates whether protein position j is a valid residue. The conformer-validity indicator $q_k \in \{0, 1\}$ indicates whether conformer slot k contains a generated geometry. Concatenation is written with square brackets; $\odot$ is element-wise multiplication; $[\ ;]$ denotes feature concatenation; σ is the sigmoid; LN denotes layer normalization; and $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse.

2.2 Section 3 equation map

Table S11: Equation-by-equation explanation for the Methodology section of the main manuscript.

Main-text label	Compact form	Main objects and dimensions	Role in the computation
dti_prediction	$s = f_\theta(D, T); p = \sigma(s); \hat{y} = \mathbf{1}[p \geq \tau]$	$s, p, \tau, \hat{y}$ are scalars	Converts a drug-target pair into a logit, probability, and thresholded prediction.
overall_pipeline	$\mathbf{D} \rightarrow \mathbf{x}_D \rightarrow \mathbf{P} \rightarrow \mathbf{P}' \rightarrow \mathbf{f} \rightarrow s$	$\mathbf{D} \in \mathbb{R}^{L_d \times d}; \mathbf{P} \in \mathbb{R}^{L_p \times d}; \mathbf{f} \in \mathbb{R}^{2d}$	Gives the order of the complete forward pass and makes the two conditioning directions explicit.
chemberta_encoding	$\mathbf{U}^{\text{chem}} = \text{ChemBERTa}(s_D)$	$\mathbf{U}^{\text{chem}} \in \mathbb{R}^{L_s \times d_c}$	Encodes the SMILES string with the frozen molecular language model.
chemberta_pooling	$\mathbf{X}_{\text{chem}} = \text{Pool}_{L_d, d}(\mathbf{U}_{\neg \text{special}}^{\text{chem}})$	$\mathbf{X}_{\text{chem}} \in \mathbb{R}^{L_d \times d}$	Removes special tokens and maps variable-length language features to the fixed drug-token interface.
gcn_projection	$\mathbf{h}_i^{g,0} = \mathbf{W}_g \mathbf{a}_i$	$\mathbf{a}_i \in \mathbb{R}^{75}; \mathbf{h}_i^{g,0} \in \mathbb{R}^d$	Projects atom descriptors into the shared hidden space; the padding indicator is prevented from adding atom semantics.
gcn_output	$\mathbf{H}^g = \text{GCN}(\mathcal{G}_D, \mathbf{H}^{g,0})$	$\mathbf{H}^g \in \mathbb{R}^{N_g \times d}$	Aggregates local bond and neighborhood information into graph tokens.
egnn_conformer	$(\mathbf{Z}_k, \mathbf{R}'_k) = \text{EGNN}(\mathbf{F}_k, \mathbf{R}_k)$	$\mathbf{F}_k \in \mathbb{R}^{N_c \times d}; \mathbf{R}_k \in \mathbb{R}^{N_c \times 3}$	Updates conformer atom features while respecting Euclidean coordinate transformations.
conformer_global	$\mathbf{c}_k = N_c^{-1} \sum_i \mathbf{z}_{k,i}$	$\mathbf{c}_k \in \mathbb{R}^d$	Produces one global summary per conformer for target-aware scoring; atom-level features are retained separately.
protein_seed_projection	$\mathbf{P}^0 = \mathbf{W}_p \text{LN}(\mathbf{P}_{\text{ProtBERT}})$	$1024 \rightarrow d; \mathbf{P}^0 \in \mathbb{R}^{L_p \times d}$	Places ProtBERT residue features in the same hidden space used by the drug branches.
protein_context	$\mathbf{x}_T = \frac{\sum_j m_j \mathbf{p}_j^0}{\max(1, \sum_j m_j)}$	$\mathbf{x}_T \in \mathbb{R}^d$	Computes the masked protein context that conditions conformer selection.
conformer_score	$a_k = \mathbf{w}_a^\top \text{SiLU}(\mathbf{W}_a[\mathbf{c}_k; \mathbf{x}_T; \mathbf{u}_k] + \mathbf{b}_a) + b_s$	$a_k \in \mathbb{R}; \mathbf{u}_k = \mathbf{W}_e \epsilon_k$	Scores each candidate geometry using its content, target context, and normalized energy.
conformer_weight	$\alpha_k = \frac{e^{a_k} q_k}{\sum_\ell e^{a_\ell} q_\ell}$	$\boldsymbol{\alpha} \in \mathbb{R}^K$	Normalizes scores over valid conformers; invalid slots receive no probability mass.
target_aware_3d	$\mathbf{Z}^{3D} = \sum_k \alpha_k \mathbf{Z}_k; \mathbf{c}^{3D} = \sum_k \alpha_k \mathbf{c}_k$	$\mathbf{Z}^{3D} \in \mathbb{R}^{N_c \times d}; \mathbf{c}^{3D} \in \mathbb{R}^d$	Forms the protein-conditioned atom-level and global 3D representations.
geo_concat	$\mathbf{X}_{\text{str}} = [\mathbf{H}^g; \mathbf{Z}^{3D}]$	$\mathbf{X}_{\text{str}} \in \mathbb{R}^{(N_g + N_c) \times d}$	Concatenates topology and target-aware geometry to match the ChemBERTa token length.
drug_gate	$\mathbf{D} = \mathbf{G}^D \odot \tilde{\mathbf{X}}_{\text{chem}} + (1 - \mathbf{G}^D) \odot \tilde{\mathbf{X}}_{\text{str}}$	$\mathbf{G}^D \in (0, 1)^{L_d \times d}$	Learns a slot- and channel-wise balance between semantic and structural drug streams.
drug_context	$\mathbf{x}_D = L_d^{-1} \sum_i \mathbf{d}_i$	$\mathbf{x}_D \in \mathbb{R}^d$	Summarizes the fused drug tokens and sends ligand context to the protein encoder.
protein_ffn_half_1	$\mathbf{U}^\ell = \frac{1}{2} \text{FFN}_1^\ell(\mathbf{P}^{\ell-1}) + \frac{1}{2} \mathbf{P}^{\ell-1}$	$\mathbf{U}^\ell \in \mathbb{R}^{L_p \times d}$	Applies a conservative first feed-forward residual update in ProteinACmix layer ℓ .

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Main-text label	Compact form	Main objects and dimensions	Role in the computation
film_params	$[\gamma^\ell, \beta^\ell] = \text{MLP}_{\text{film}}^\ell(\mathbf{x}_D)$	$\gamma^\ell, \beta^\ell \in \mathbb{R}^d$	Generates ligand-dependent feature-wise scale and shift parameters; $\rho = 0.05$ limits their initial effect.
ligand_film	$\tilde{\mathbf{U}}^\ell = \mathbf{U}^\ell \odot (1 + \tilde{\gamma}^\ell) + \tilde{\beta}^\ell$	Broadcast over L_p residue tokens	Modulates protein features with the fused ligand context.
protein_acmix	$\mathbf{A}^\ell = \text{MHSA}_{\text{rel}}(\tilde{\mathbf{U}}^\ell) + \tilde{\mathbf{U}}^\ell; \mathbf{C}^\ell = \text{Conv}(\mathbf{A}^\ell) + \mathbf{A}^\ell$	All tensors are $L_p \times d$	Combines long-range relative attention, local convolution, and a second half-step FFN.
protein_output	$\mathbf{P} = \mathbf{P}^3$	$\mathbf{P} \in \mathbb{R}^{L_p \times d}$	Defines the ligand-conditioned protein sequence passed to refinement.
feature_projection	$\psi_{\text{chem}} : \mathbb{R}^{128} \rightarrow \mathbb{R}^{128}; \psi_g : \mathbb{R}^{75} \rightarrow \mathbb{R}^{128}; \psi_p : \mathbb{R}^{1024} \rightarrow \mathbb{R}^{128}$	Three active input mappings	Records the shared-dimension contract between cached features and trainable encoders.
feature_contexts	$\mathbf{x}_T = \text{MaskedMean}(\mathbf{P}^0, \mathbf{m}); \mathbf{x}_D = \text{MeanPool}(\mathbf{D})$	Two vectors in $\mathbb{R}^d$	Documents the order of the two feedback paths: target-to-drug first, then drug-to-protein.
protein_masked_input	$\mathbf{P}_j^{\text{mask}} = m_j \mathbf{P}_j$	$\mathbf{P}^{\text{mask}} \in \mathbb{R}^{L_p \times d}$	Suppresses padded residue positions before protein-side Mamba processing.
bidirectional_mamba	$\mathbf{M} = \mathbf{W}_m[\mathbf{M}_f; \mathbf{M}_b]$	$\mathbf{M}_f$ and $\mathbf{M}_b$ are forward/backward Mamba streams	Captures ordered protein dependencies in both sequence directions; the backward stream is computed on the reversed sequence.
mamba_residual	$\mathbf{R} = \mathbf{P}^{\text{mask}} + \text{Dropout}(\mathbf{M}); \mathbf{P}_{\text{Mamba}} = \mathbf{R} + \text{FFN}_m(\mathbf{R})$	$L_p \times d$	Adds residual and feed-forward refinement, then reapplies the residue mask.
mamba_gate	$g = \sigma(\eta); \mathbf{P}' = \mathbf{P} + g(\mathbf{P}_{\text{Mamba}} - \mathbf{P})$	$g \in (0, 1); \mathbf{P}' \in \mathbb{R}^{L_p \times d}$	Controls the strength of Mamba injection; $\eta = -2$ initializes $g \approx 0.119$.
bi_self_attention	$\mathbf{D}^s = \text{SA}_D(\mathbf{D}); \mathbf{P}^s = \text{SA}_P(\mathbf{P}')$	Drug and protein streams retain their lengths	Refines each modality independently before cross-modal intention updates.
intention_head	$\mathbf{Q}_r = \mathbf{Q}\mathbf{W}_r^Q; \mathbf{K}_r = \mathbf{X}\mathbf{W}_r^K; \mathbf{V}_r = \mathbf{X}\mathbf{W}_r^V$	$d_h = d/H$ per head	Builds a feature-space interaction map using a ridge-stabilized pseudoinverse and head weighting.
intention_block	$\Phi(\mathbf{X}, \mathbf{Q}) = \beta \mathbf{Q}^\top \text{softmax}(\text{Int}(\text{LN}(\mathbf{X}), \mathbf{Q}))$	$\Phi \in \mathbb{R}^{d \times d}$	Converts cross-modal intention features into a compact pair representation.
bidirectional_intention_update	$\mathbf{V}_p^\ell = \Phi(\mathbf{D}^{\ell-1}, \mathbf{P}^{\ell-1}); \mathbf{V}_d^\ell = \Phi(\mathbf{P}^{\ell-1}, \mathbf{D}^{\ell-1})$	Drug and protein directions are computed separately	Keeps the interaction update bidirectional rather than concatenating unimodal features.
fusion_pooling	$\mathbf{f} = [\text{MaxPool}(\mathbf{D}^{(L)}); \text{MaxPool}(\mathbf{P}^{(L)})]$	$\mathbf{f} \in \mathbb{R}^{2d} = \mathbb{R}^{256}$	Produces the fixed-size pair vector consumed by the prediction head.
prediction_mlp	$\mathbf{f} \rightarrow \mathbf{z}_1 \rightarrow \mathbf{z}_2 \rightarrow \mathbf{z}_3 \rightarrow s$	$256 \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow 1$	Maps the pair vector to one interaction logit using ReLU and layer normalization.
weighted_bce	$\mathcal{L} = -B^{-1} \sum_i [w_+ y_i \log p_i + (1 - y_i) \log(1 - p_i)]$	B samples; $w_+ > 0$	Optimizes the logit-derived probability while compensating for positive/negative imbalance.
pos_weight	$w_+ = N_- / N_+$	Training-split counts only	Computes the dataset-specific positive-class weight without using validation or test labels.

2.3 Interpretation of the conditioning order

The key implementation contract is the order of operations. ProtBERT features are projected and masked to form $\mathbf{x}_T$, which scores and aggregates the candidate conformers. The resulting fused drug sequence is pooled to form $\mathbf{x}_D$, which then modulates the protein sequence through FiLM. Mamba is applied only to the ordered protein stream, after ligand conditioning and before BiIntention. This order should be preserved when reproducing the model; moving the two context vectors or applying the Mamba block to the drug slots changes the stated architecture.

The conformer energy ϵ_k is an auxiliary ranking feature. It is not treated as a calibrated thermodynamic free energy and is not compared across molecules as an absolute physical quantity. Invalid conformers are masked in the softmax, and the implementation activates the first slot as a fallback if all conformer-validity indicators are zero.

3 Standalone Tables

The following tables are supplied as standalone, selectable-text versions for reviewer inspection and high-resolution reading. They reproduce the numerical values reported in the manuscript and add no new statistical claims. The equation guide and the machine-readable data/code links in the manuscript should be used for interpretation and reuse.

Table S12: Dataset statistics used in the main experiments. Counts are computed from the processed training, validation, and test split files used in this study.

Dataset	Drugs	Proteins	Interactions	Positive	Negative	Positive rate
BindingDB	14,643	2,623	49,199	20,674	28,525	42.02%
BioSNAP	4,505	2,181	27,464	13,830	13,634	50.36%
Human	2,726	2,001	5,997	2,633	3,364	43.91%
C. elegans	1,767	1,876	7,786	3,893	3,893	50.00%

Table S13: Basic training hyperparameters for the main experiments.

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Maximum epochs	100
Micro-batch size	16 per GPU
Gradient accumulation	4 steps
Gradient clipping	1.0
Learning-rate decay factor	0.5
Minimum learning rate	1×10^{-6}

Table S14: Dataset-specific positive-class weights used by the binary cross-entropy loss.

Dataset	Training positives	Training negatives	Positive-class weight
BindingDB	14,583	19,856	1.3616
BioSNAP	9,684	9,540	0.9851
Human	1,846	2,351	1.2736
C. elegans	2,727	2,723	0.9985

Table S15: AUROC/AUPRC comparison on four benchmark datasets. Baseline values are adopted from reported results; TAMR-DTI values are from five-seed runs and are shown as mean $\pm$ standard deviation.

Method	BindingDB		BioSNAP		Human		C. elegans	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
SVM	0.904 $\pm$ 0.002	0.865 $\pm$ 0.001	0.819 $\pm$ 0.045	0.839 $\pm$ 0.038	0.913 $\pm$ 0.012	0.905 $\pm$ 0.001	0.925 $\pm$ 0.009	0.907 $\pm$ 0.002
RF	0.942 $\pm$ 0.001	0.923 $\pm$ 0.001	0.857 $\pm$ 0.009	0.872 $\pm$ 0.006	0.939 $\pm$ 0.009	0.927 $\pm$ 0.001	0.935 $\pm$ 0.009	0.931 $\pm$ 0.001
KNN	0.895 $\pm$ 0.001	0.865 $\pm$ 0.002	0.842 $\pm$ 0.003	0.805 $\pm$ 0.003	0.915 $\pm$ 0.002	0.902 $\pm$ 0.002	0.912 $\pm$ 0.003	0.896 $\pm$ 0.002
LR	0.916 $\pm$ 0.003	0.884 $\pm$ 0.002	0.821 $\pm$ 0.004	0.796 $\pm$ 0.002	0.895 $\pm$ 0.003	0.862 $\pm$ 0.001	0.892 $\pm$ 0.002	0.883 $\pm$ 0.003
GraphDTA	0.944 $\pm$ 0.004	0.925 $\pm$ 0.006	0.871 $\pm$ 0.004	0.870 $\pm$ 0.006	0.965 $\pm$ 0.003	0.955 $\pm$ 0.003	0.959 $\pm$ 0.004	0.951 $\pm$ 0.002
DrugBAN	0.952 $\pm$ 0.001	0.936 $\pm$ 0.001	0.902 $\pm$ 0.001	0.905 $\pm$ 0.002	0.979 $\pm$ 0.001	0.969 $\pm$ 0.005	0.981 $\pm$ 0.002	0.978 $\pm$ 0.002
IIFDTI	0.924 $\pm$ 0.002	0.901 $\pm$ 0.004	0.895 $\pm$ 0.003	0.898 $\pm$ 0.005	0.977 $\pm$ 0.003	0.967 $\pm$ 0.027	0.984 $\pm$ 0.002	0.979 $\pm$ 0.002
TransformerCPI	0.895 $\pm$ 0.002	0.861 $\pm$ 0.003	0.843 $\pm$ 0.008	0.859 $\pm$ 0.007	0.956 $\pm$ 0.005	0.943 $\pm$ 0.002	0.975 $\pm$ 0.003	0.975 $\pm$ 0.002
MolTrans	0.935 $\pm$ 0.003	0.901 $\pm$ 0.004	0.879 $\pm$ 0.006	0.882 $\pm$ 0.006	0.979 $\pm$ 0.003	0.975 $\pm$ 0.002	0.982 $\pm$ 0.002	0.979 $\pm$ 0.002
BINDTI	0.956 $\pm$ 0.001	0.942 $\pm$ 0.002	0.895 $\pm$ 0.003	0.894 $\pm$ 0.003	0.980 $\pm$ 0.001	0.975 $\pm$ 0.004	0.985 $\pm$ 0.002	0.984 $\pm$ 0.002
LDM-DTI	0.960 $\pm$ 0.002	0.945 $\pm$ 0.002	0.908 $\pm$ 0.003	0.906 $\pm$ 0.003	0.981 $\pm$ 0.001	0.976 $\pm$ 0.002	0.988 $\pm$ 0.002	0.986 $\pm$ 0.002
TAMR-DTI	0.964 $\pm$ 0.001	0.954 $\pm$ 0.002	0.930 $\pm$ 0.002	0.935 $\pm$ 0.002	0.982 $\pm$ 0.002	0.978 $\pm$ 0.002	0.992 $\pm$ 0.001	0.991 $\pm$ 0.001

Table S16: TAMR-DTI repeated-run summary across the four benchmark datasets. Values are mean $\pm$ standard deviation over seeds 42, 43, 44, 45, and 46.

Dataset	AUROC	AUPRC	Acc@0.5	F1@0.5
BindingDB	0.964 $\pm$ 0.001	0.954 $\pm$ 0.002	0.909 $\pm$ 0.003	0.891 $\pm$ 0.004
BioSNAP	0.930 $\pm$ 0.002	0.935 $\pm$ 0.002	0.857 $\pm$ 0.004	0.859 $\pm$ 0.004
Human	0.982 $\pm$ 0.002	0.978 $\pm$ 0.002	0.930 $\pm$ 0.009	0.924 $\pm$ 0.009
C. elegans	0.992 $\pm$ 0.001	0.991 $\pm$ 0.001	0.961 $\pm$ 0.004	0.961 $\pm$ 0.004

Table S17: Single-seed diagnostic ablation results on BioSNAP. All variants use seed 42, the same split, validation-AUROC checkpoint selection, and fixed-threshold Acc@0.5/F1@0.5.

Variant	AUROC	AUPRC	Acc@0.5	F1@0.5
Full TAMR-DTI	0.933	0.937	0.862	0.864
w/o Target-aware conformer	0.916	0.922	0.844	0.845
w/o Protein FiLM	0.920	0.924	0.842	0.844
w/o Protein Mamba refinement	0.918	0.925	0.842	0.845
w/o Bidirectional modulation	0.920	0.926	0.848	0.849

Table S18: Single-seed conformer-count comparison on BioSNAP. Conformers are energy-sorted during preprocessing; $n = 1$ keeps the lowest-energy conformer only, while larger n values expose additional geometries.

Variant	Effective conformers	Aggregation	Best epoch	AUROC	AUPRC	Acc@0.5	F1@0.5
$n = 1$ target-aware	1	Target-aware	46	0.9277	0.9297	0.8578	0.8591
$n = 2$ target-aware	2	Target-aware	68	0.9300	0.9322	0.8575	0.8605
$n = 4$ target-aware	4	Target-aware	46	0.9297	0.9345	0.8524	0.8542
$n = 8$ target-aware	8	Target-aware	51	0.9322	0.9356	0.8575	0.8624
$n = 16$ target-aware	16	Target-aware	62	0.9312	0.9326	0.8631	0.8668
$n = 8$ uniform average	8	Uniform average	64	0.9308	0.9340	0.8589	0.8616

4 High-Resolution Figures

The figures below are included directly from the original vector PDF artwork in the manuscript source. Text, lines, and plotted curves therefore remain selectable and are not rasterized by this SI compilation. The figure numbering is local to the SI (Figures S1–S7); the corresponding main-text figure labels are given in each caption.

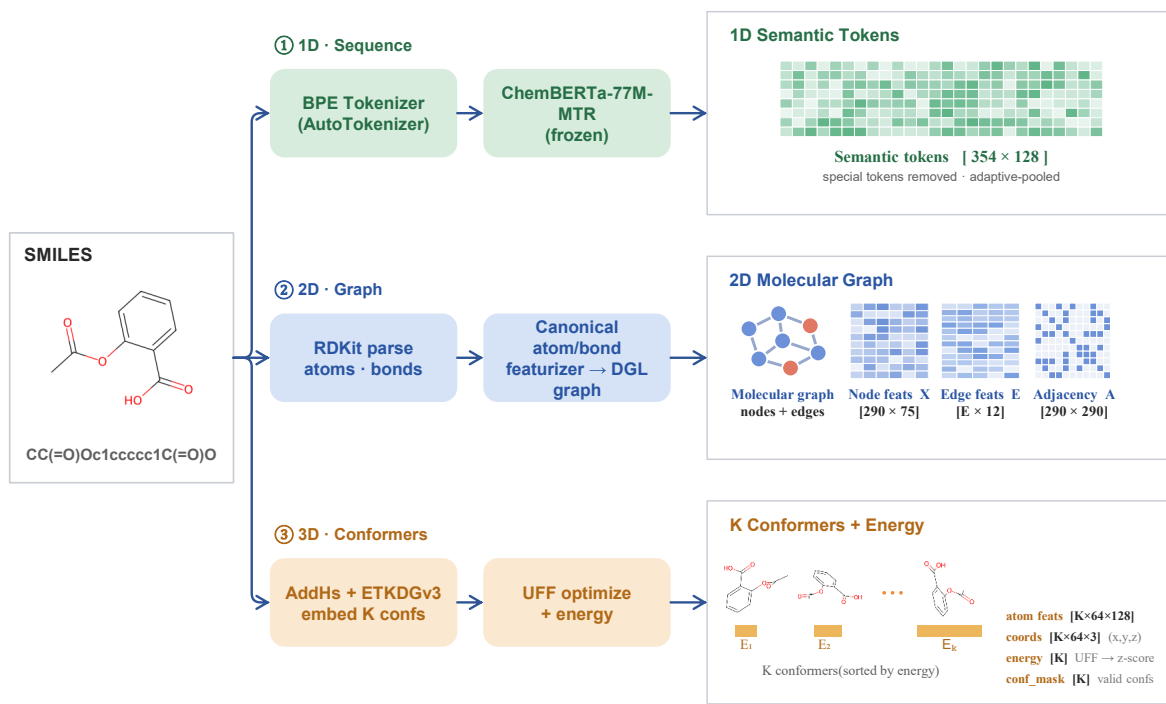


Figure S1: High-resolution version of main-text Figure 1: drug data preprocessing. Starting from a SMILES string, TAMR-DTI constructs a ChemBERTa semantic-token matrix, an RDKit/DGL molecular graph, and a set of energy-ranked 3D conformers with atom features, coordinates, energies, and conformer-validity indicators.

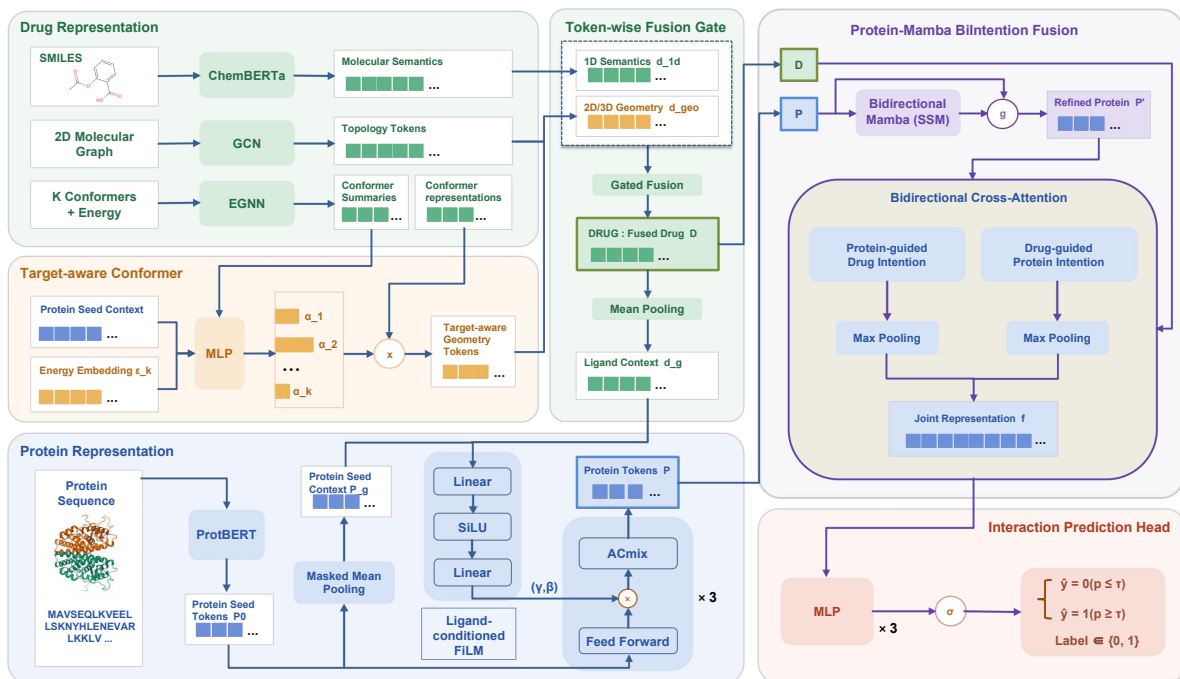


Figure S2: High-resolution version of main-text Figure 2: TAMR-DTI representation framework. The artwork shows the drug representation streams, ligand-conditioned protein representation module, protein-side refinement block, and BiIntention interaction head.

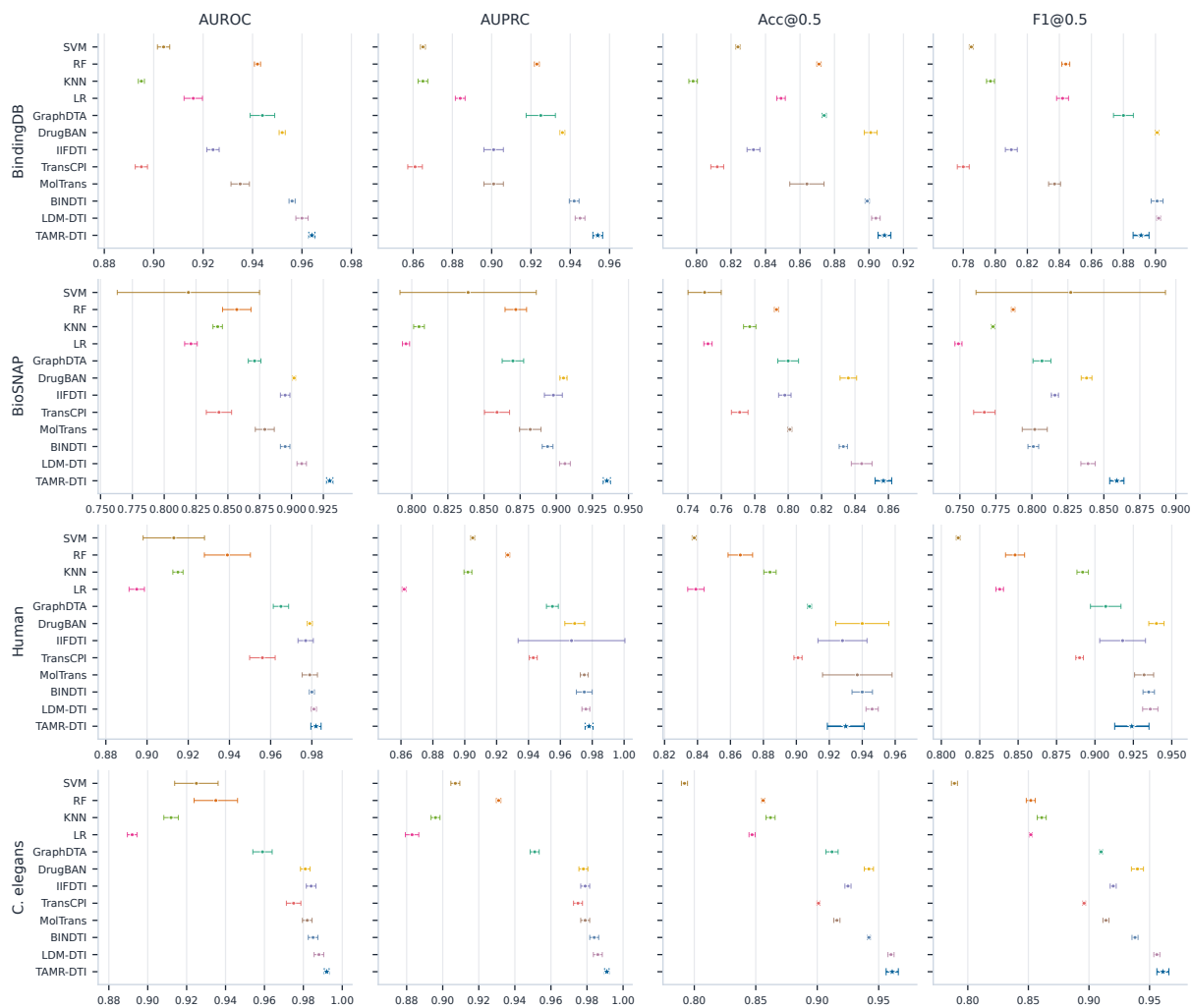


Figure S3: High-resolution version of the main-text all-method comparison figure. The plot shows metric values and uncertainty intervals for AUROC, AUPRC, Acc@0.5, and F1@0.5 across the four benchmark datasets.

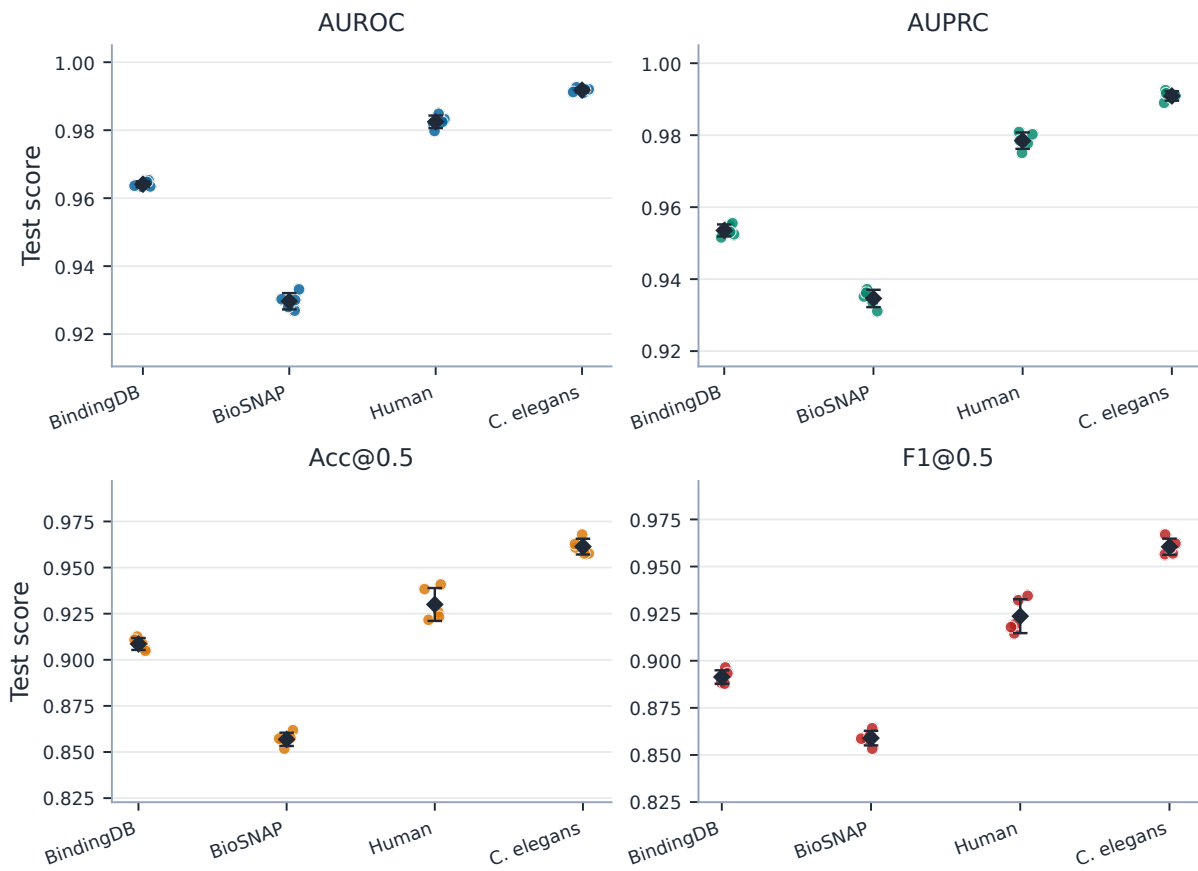


Figure S4: High-resolution version of the main-text seed-stability figure. Each point represents one fixed-seed run and the intervals summarize the repeated-run variation under the stated evaluation protocol.

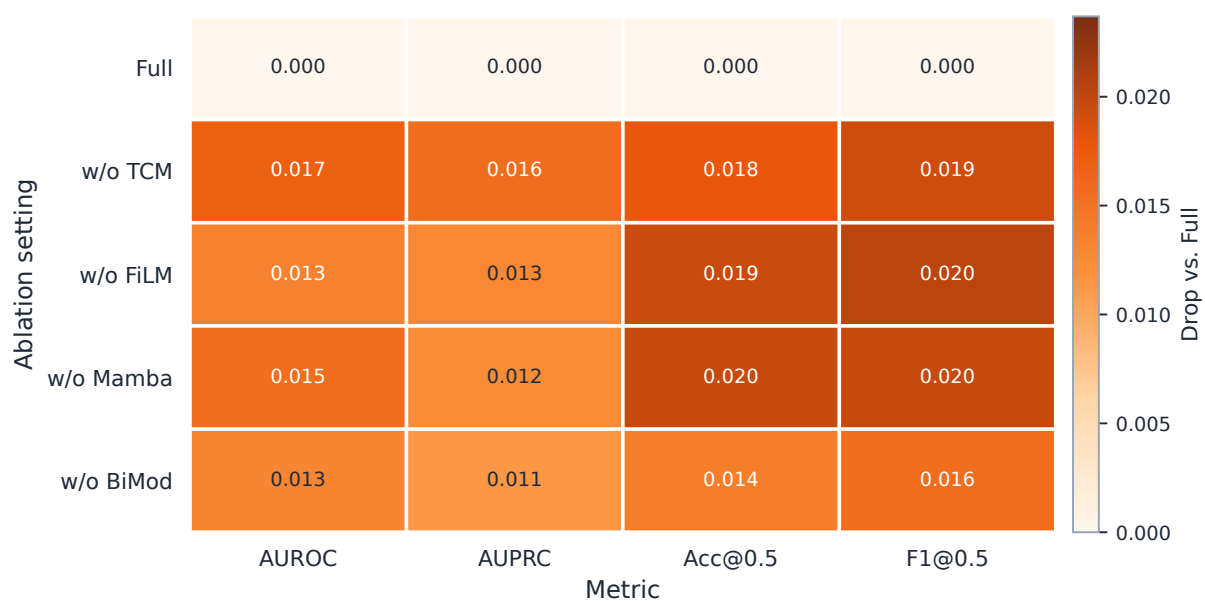


Figure S5: High-resolution version of the main-text BioSNAP ablation-drop figure. Each cell reports the relative test-score decrease caused by removing one module under the same seed, split, checkpoint selection rule, and fixed threshold; the complete model is the zero-drop reference.

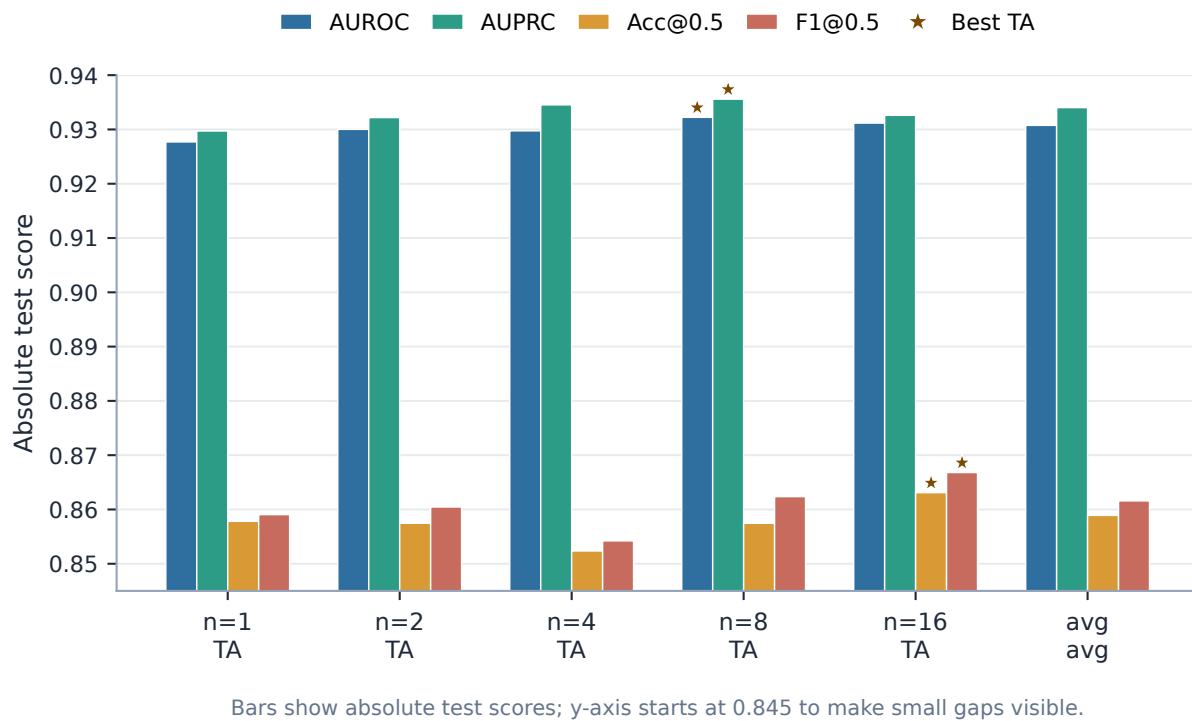


Figure S6: High-resolution version of the main-text conformer-count comparison figure. The plot compares effective conformer counts and uniform versus target-aware aggregation on BioSNAP under the single-seed diagnostic protocol.

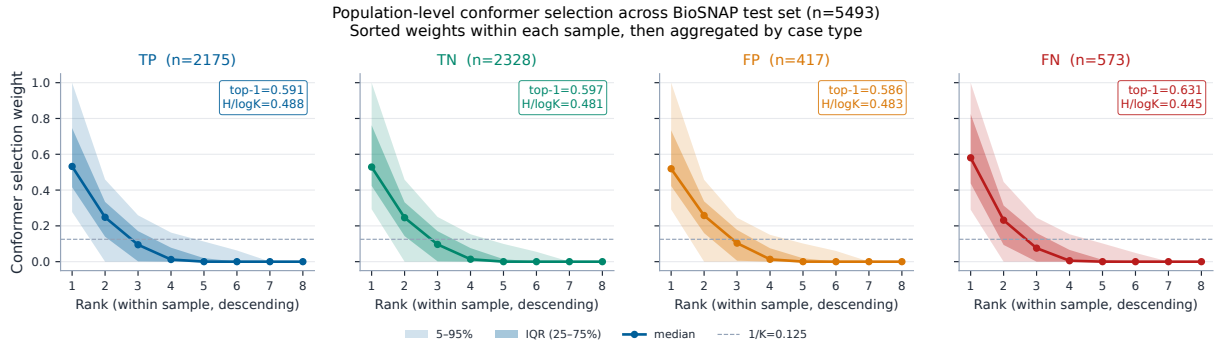


Figure S7: High-resolution version of the main-text interpretability figure. For each BioSNAP test pair, the eight conformer weights are sorted within the sample and summarized by prediction group; solid curves are medians, dark bands are interquartile ranges, and light bands are the 5–95% ranges.